CORE PURE / COMPLEX NUMBERS

Complex Roots and Geometry Problems

Mark scheme

13 questions · 127 marks

Questions

Question	Marks by part	Marks	Page
Question 1	(a)(i) 2 (a)(ii) 2 (b) 4	8	4
Question 2	(a) 5 (b) 2 (c) 4	11	6
Question 3	(a) 4 (b) 5	9	8
Question 4	(a) 2 (b) 2 (c) 2 (d) 2 (e) 2	10	9
Question 5		5	11
Question 6	(a) 6 (b) 4	10	12
Question 7	(a) 3 (b) 4	7	13
Question 8	(a) 2 (b) 1 (c) 2 (d) 3 (e) 4	12	14
Question 9	(a) 2 (b) 2 (c)(i) 1 (c)(ii) 4	9	16
Question 10	(i) 5 (ii) 2 (iii) 3 (iv) 3	13	17
Question 11	(a)(i) 1 (a)(ii) 3 (a)(iii) 3 (b)(i) 2 (b)(ii) 2 (b)(iii) 3	14	19
Question 12	(a) 2 (b) 4 (c) 2	8	21
Question 13	(i) 2 (ii) 6 (iii) 1 (iv) 2	11	22
	Total	**127**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

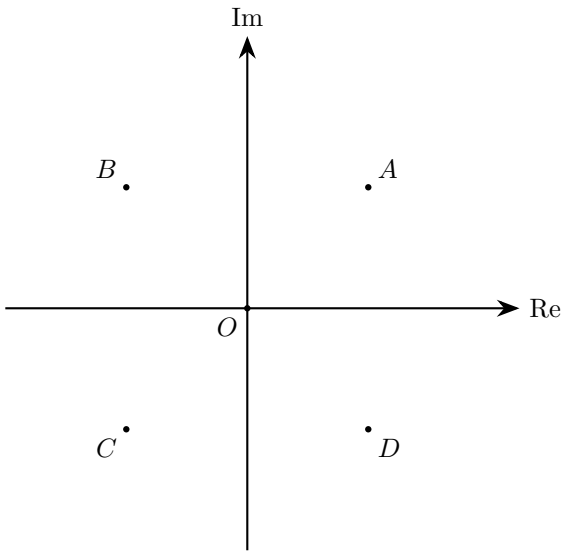
- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

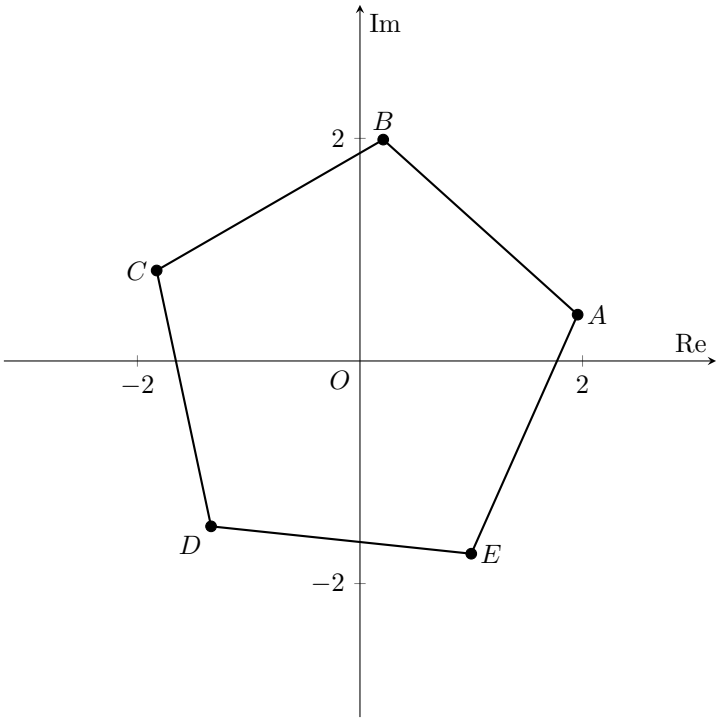
Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)(i)			
	Labelled point	**B1**	B and D at the quarter-turn rotations of A about O, B anticlockwise from A. Condone small inaccuracy in position.
	Symmetry	**B1**	C opposite A through O, labelled C.
		[2]	
1(a)(ii)	$z_1 + z_1 e^{\frac{1}{2}\pi i} + z_1 e^{\pi i} + z_1 e^{\frac{3}{2}\pi i}$ $= z_1 + iz_1 - z_1 - iz_1$	**M1**	Evaluates the three exponentials, directly or by $\cos\theta + i\sin\theta$, at least two correctly, and sums the resulting multiples of z_1.
	$= z_1(1 + i - 1 - i) = 0$	**A1 AG**	All three values correct and the terms shown to cancel to 0 without error.
	Alternative: Geometric series		
	$z_1\left(1 + e^{\frac{1}{2}\pi i} + e^{\pi i} + e^{\frac{3}{2}\pi i}\right)$ $= z_1 \dfrac{1 - \left(e^{\frac{1}{2}\pi i}\right)^4}{1 - e^{\frac{1}{2}\pi i}}$	**M1**	Uses the sum of four terms of a geometric series with ratio $e^{\frac{1}{2}\pi i}$. Condone the factor z_1 omitted here.
	$\left(e^{\frac{1}{2}\pi i}\right)^4 = e^{2\pi i} = 1 \Rightarrow$ sum $= 0$	**A1 AG**	Shows the ratio to the fourth power is 1 and concludes the sum is 0, with the denominator nonzero.
		[2]	
1(b)	$-16 = 16e^{\pi i}$ $= 16e^{(\pi + 2k\pi)i}$	**B1**	Accept modulus–argument form, and the four arguments listed in place of $\pi + 2k\pi$.

Question 1 continued

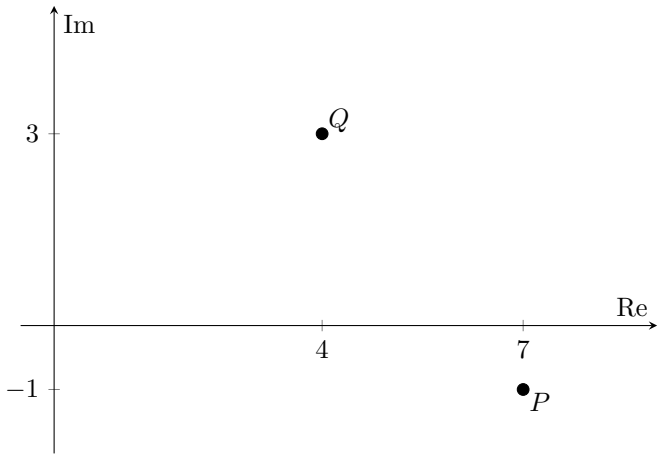
Question	Scheme	Marks	Notes
	$z = 2e^{\frac{\pi+2k\pi}{4}i}$, $k = 0, 1, 2, 3$	**M1**	Takes the fourth root of their modulus and divides their arguments by 4, giving at least two distinct roots in any form.
	$z = \sqrt{2} + \sqrt{2}\,i,\ -\sqrt{2} + \sqrt{2}\,i$	**A1**	Any two correct roots, in any form.
	$z = \pm\sqrt{2} \pm \sqrt{2}\,i$ (all four)	**A1**	All four in the form $a + bi$ with exact a and b, and no others. Roots left in exponential form score A0.
		[4]	
Total for Question 1		**8**	

Question	Scheme	Marks	Notes
2(a)	$\lvert w\rvert = \sqrt{16^2 + (16\sqrt{3})^2}$ $\arg w = \arctan \frac{16\sqrt{3}}{16}$	**M1**	Attempts both the modulus and the argument of $16 + 16\sqrt{3}\,i$. Condone arithmetic slips.
	$16 + 16\sqrt{3}\,i = 32e^{\frac{\pi}{3}i}$	**A1**	Or modulus 32 and argument $\frac{\pi}{3}$ (awrt 1.05) stated.
	$z =$ "32"$^{\frac{1}{5}}\, e^{\frac{\text{"}\pi/3\text{"}+2k\pi}{5}i}$	**M1**	Takes the fifth root of their modulus and uses arguments (their argument $+2k\pi$)/5, for general k or at least two values, or rotates one root by $\frac{2\pi}{5}$.
	$2e^{\frac{\pi}{15}i},\ 2e^{\frac{7\pi}{15}i}, \ldots$	**A1**	Any two correct roots, in any form.
	$2e^{\frac{\pi}{15}i},\ 2e^{\frac{7\pi}{15}i},\ 2e^{\frac{13\pi}{15}i},$ $2e^{\frac{19\pi}{15}i},\ 2e^{\frac{5\pi}{3}i}$	**A1**	All five as $2e^{i\theta}$ with $0 \le \theta < 2\pi$, exact or awrt 0.209, 1.47, 2.72, 3.98, 5.24, and no others. A0 for any argument outside the range.
		[5]	
2(b)			
	Shape	**B1**	Pentagon centred at O with vertices equally spaced on a circle of radius 2, the radius shown by a scale or label.

Question 2 continued

Question	Scheme	Marks	Notes
	Labelled point	**B1**	Each vertex labelled with its root, or a letter identified with it, in its correct position, one near the positive real axis in the first quadrant.
		[2]	
2(c)	Five congruent triangles with sides $2, 2$ and angle $\frac{2\pi}{5}$ at O	**B1**	Angle at O may be implied by its use in an area.
	$\frac{1}{2} \times 2 \times 2 \times \sin \frac{2\pi}{5}$	**M1**	Uses $\frac{1}{2}ab\sin C$ with two sides equal to their modulus and their angle at O as the included angle.
	$= 2\sin \frac{2\pi}{5}$	**A1**	
	$5 \times 2\sin \frac{2\pi}{5} = 10\sin \frac{2\pi}{5} \Rightarrow k = 10$	**A1**	cso. Area $10\sin \frac{2\pi}{5}$ or $k = 10$ stated; a decimal area alone scores A0.
		[4]	
	Total for Question 2	**11**	

Question	Scheme	Marks	Notes
3(a)	$(h-1)^2+4^2=(h-7)^2+2^2$ $\Rightarrow h=\ldots$	**M1**	Equates the distances, or squared distances, from $(h,0)$ to A and B, or intersects the perpendicular bisector of AB with the real axis. Condone sign slips.
	$h=3$	**A1**	
	$r^2=(\text{“}3\text{”}-1)^2+4^2$ $[=20]$	**M1**	Uses Pythagoras for the distance from their real centre to A or B.
	$\lvert z-3\rvert=2\sqrt{5}$	**A1**	Accept $\sqrt{20}$ or awrt 4.47. Only the form $\lvert z-a\rvert=b$ scores; $(x-3)^2+y^2=20$ alone scores A0.
		[4]	
3(b)	$(x-3)^2+y^2=20$ $(x-1)^2+(y-1)^2=25$	**M1 A1**	**M1** Writes at least one circle as an equation in x and y, using $z=x+iy$ and their C of the form $\lvert z-a\rvert=b$. Condone sign slips. **A1** Both correct, in any form.
	$4x-2y-12=0\Rightarrow y=2x-6$	**M1**	Subtracts their equations to eliminate x^2 and y^2, reaching a linear equation in x and y. Condone sign slips.
	$(x-3)^2+(\text{“}2x-6\text{”})^2=20$ $\Rightarrow x=\ldots$	**dM1**	Substitutes their linear equation into either circle and solves the resulting quadratic in one variable by any method.
	$z_1=1-4i,\ z_2=5+4i$	**A1**	Both, in either order, as complex numbers; coordinates alone score A0.
		[5]	
	Total for Question 3	**9**	

Question	Scheme	Marks	Notes
4(a)	$z_2 = \frac{(7-i)(1+i)}{(1-i)(1+i)} = \ldots$	**M1**	Multiplies numerator and denominator by $1+i$ and expands, or equates real and imaginary parts of $(a+bi)(1-i) = 7-i$. Condone one slip.
	$z_2 = 4+3i$	**A1**	
		[2]	
4(b)			
	Labelled point	**B1**	P plotted and labelled at $(7,-1)$, in the fourth quadrant, with its position shown by scale or coordinates.
	Labelled point	**B1ft**	ft their z_2 from (a). Q plotted and labelled at their point, positioned correctly relative to P.
		[2]	
4(c)	$\overrightarrow{QO} \cdot \overrightarrow{QP} = (-4)(3) + (-3)(-4)$	**M1**	Applies a perpendicularity test at Q to OQ and QP, such as a scalar product, a product of gradients or Pythagoras' converse. Condone sign slips.
	$= 0 \Rightarrow \angle OQP = \frac{\pi}{2}$	**A1 AG**	Correct values giving 0, a gradient product of -1, or $OQ^2 + QP^2 = OP^2$, with a conclusion that the angle is a right angle.
		[2]	
4(d)	Uses OP as a diameter: $C = \frac{1}{2}(7-i) = \ldots$	**M1**	Takes C as the midpoint of OP, or solves for the point equidistant from O, P and Q.

Question 4 continued

Question	Scheme	Marks	Notes
	$C = \frac{7}{2} - \frac{1}{2}i$	**A1**	oe as a complex number, such as $3.5 - 0.5i$. Coordinates $\left(\frac{7}{2}, -\frac{1}{2}\right)$ alone do not score.
		[2]	
4(e)	$r = \frac{1}{2}\lvert OP\rvert = \frac{1}{2}\sqrt{7^2 + (-1)^2}$	**M1**	Finds half of $\lvert OP\rvert$, or the distance from their C to O, P or Q, by Pythagoras. Condone sign slips.
	$r = \frac{5\sqrt{2}}{2}$	**A1**	Any exact form, such as $\frac{\sqrt{50}}{2}$ or $\sqrt{12.5}$.
		[2]	
	Total for Question 4	**10**	

Question	Scheme	Marks	Notes
5	A, B, C: $z^3 - 1 = 0$	**B1**	
	P, Q, R: $(z-2)^3 = 1$	**M1**	Replaces z by $z-2$ in their cubic, or forms a cubic from the three roots by factors or root sums and products. Need not be expanded.
	$z^3 - 6z^2 + 12z - 9 = 0$	**A1**	Correct cubic, oe such as $(z-3)(z^2-3z+3) = 0$.
	$(z^3-1)(z^3-6z^2+12z-9) = 0$ $\Rightarrow z^6 - 6z^5 + \ldots = 0$	**M1**	Multiplies their two cubics and expands. Condone slips in collecting terms.
	$z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9 = 0$	**A1**	Any nonzero integer multiple accepted.
	Total for Question 5	**5**	

Question	Scheme	Marks	Notes
6(a)	$z_A - z_G = 4 + 4i$	**B1**	
	$z_B - z_G = (4+4i)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ $= \ldots$	**M1**	Multiplies their $z_A - z_G$ by $e^{\pm 2\pi i/3}$ in any form and expands. Condone a sign slip in the cosine or sine value.
	$z_B - z_G = (-2 - 2\sqrt{3}) + (2\sqrt{3} - 2)i$	**A1**	
	$z_B = (-1 - 2\sqrt{3}) + (2\sqrt{3} - 4)i$	**A1**	Exact, in the form $a + bi$. B and C may be interchanged.
	$z_C = (1 - 2i) + (4 + 4i)\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$ $= \ldots$	**M1**	Rotates their $z_A - z_G$ through the other angle and adds z_G, or uses $z_C = 3z_G - z_A - z_B$ with their z_B.
	$z_C = (-1 + 2\sqrt{3}) - (4 + 2\sqrt{3})i$	**A1**	Exact, in the form $a + bi$.
		[6]	
6(b)	$z_D = \frac{1}{3}(z_G + z_B + z_C)$, $z_B + z_C = 3z_G - z_A$ $\Rightarrow z_D = \frac{1}{3}(4z_G - z_A)$	**M1**	Writes z_D as the mean of z_G, z_B, z_C, then simplifies using $z_A + z_B + z_C = 3z_G$ or evaluates with their (a) values.
	$z_D - z_G = -\frac{1}{3}(z_A - z_G)$, similarly for E, F $\Rightarrow$ area factor $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$	**A1**	Lengths scaled by $\frac{1}{3}$ about G, sign not needed, with E and F by symmetry, and area factor $\frac{1}{9}$.
	$GA = \lvert 4 + 4i \rvert = 4\sqrt{2}$ Area $ABC = 3 \times \frac{1}{2}(4\sqrt{2})^2 \sin\frac{2\pi}{3}$ $= \ldots$ $[= 24\sqrt{3}]$	**M1**	Uses three triangles $\frac{1}{2}r^2 \sin\frac{2\pi}{3}$ with their $r = \lvert z_A - z_G \rvert$, or $\frac{\sqrt{3}}{4}s^2$ with their side $s = \lvert z_A - z_B \rvert$.
	Area $DEF = \frac{1}{9} \times 24\sqrt{3} = \frac{8\sqrt{3}}{3}$	**A1**	Exact, oe such as $\frac{8}{\sqrt{3}}$.
		[4]	
	Total for Question 6	**10**	

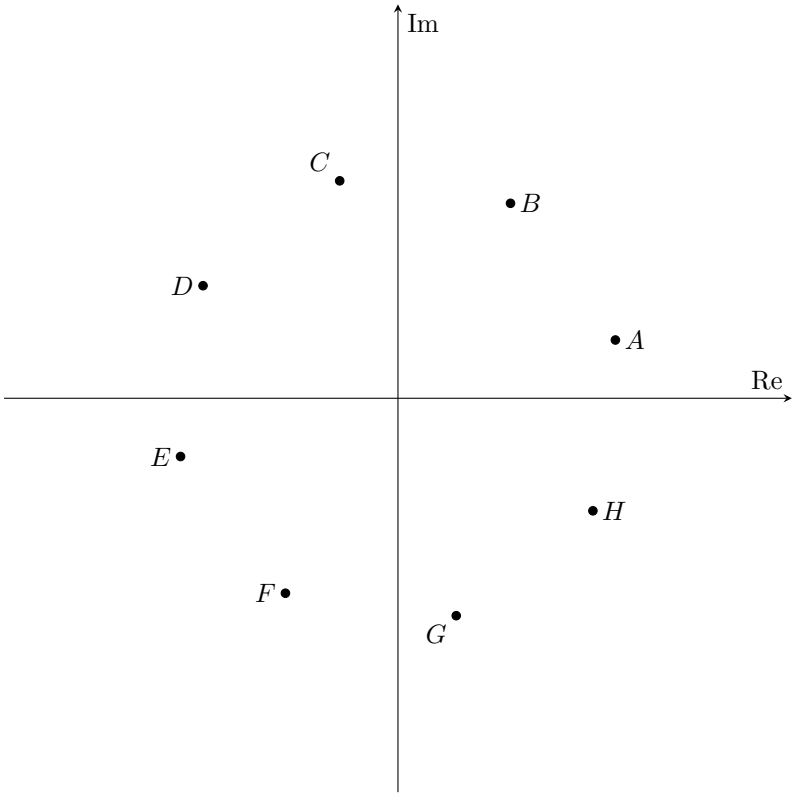
Question	Scheme	Marks	Notes
7(a)	$\lvert w-(1+i)\rvert^2 = \big(w-(1+i)\big)\big(w^*-(1-i)\big)$ $= \ldots$	**M1**	Multiplies $w-(1+i)$ by its attempted conjugate and expands, or an equivalent Cartesian method. Condone sign slips.
	$= ww^* - (1-i)w - (1+i)w^* + (1+i)(1-i)$	**A1**	Correct four-term expansion, oe.
	$ww^* = \lvert w\rvert^2 = 1$, $(1+i)(1-i) = 2$ $\Rightarrow \lvert w-(1+i)\rvert^2 = 3-(1-i)w-(1+i)w^*$	**A1 AG**	Uses $ww^* = 1$ from $\lvert w\rvert = 1$ and $(1+i)(1-i) = 2$, reaching the printed result without error.
		[3]	
7(b)	A_{2r-1}: $w_r = e^{\frac{2(r-1)\pi}{5}i}$, the fifth roots of unity P: $1+i$	**B1**	Needs both the odd-numbered vertices as the fifth roots of unity, or their five arguments, and P as $1+i$.
	$\sum_{r=1}^{5}(PA_{2r-1})^2 = \sum_{r=1}^{5}\big(3-(1-i)w_r-(1+i)w_r^*\big)$ $= 15-(1-i)\sum w_r-(1+i)\sum w_r^*$	**M1**	Sums the result of (a) over the five odd-numbered vertices, separating the sums of w_r and w_r^*. Condone the constant left as $\sum 3$.
	$\sum w_r = 0$, $\sum w_r^* = 0$	**B1**	Both sums stated or shown to be zero.
	$\sum_{r=1}^{5}(PA_{2r-1})^2 = 15$	**A1**	cso.
		[4]	
	Total for Question 7	**7**	

Question	Scheme	Marks	Notes
8(a)	$\alpha^5 = \cos\left(5 \times \frac{3\pi}{5}\right) + i\sin\left(5 \times \frac{3\pi}{5}\right)$ $= \cos 3\pi + i\sin 3\pi$	**M1**	Applies de Moivre's theorem to α^5, reaching argument 3π or an equivalent.
	$\alpha^5 = -1$, so α is a root	**A1 AG**	Evaluates $\cos 3\pi = -1$ and $\sin 3\pi = 0$ and concludes that α is a root.
		[2]	
8(b)	$\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$, -1, $\cos\frac{7\pi}{5} + i\sin\frac{7\pi}{5}$, $\cos\frac{9\pi}{5} + i\sin\frac{9\pi}{5}$	**B1**	All four, in any correct form, with arguments in any range such as $-\pi < \theta \leq \pi$.
		[1]	
8(c)	$\alpha^5 + 1 = (\alpha+1)(\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1)$	**M1**	Factorises $z^5 + 1$ with factor $z + 1$, or sums the geometric series with ratio $-\alpha$. Expanding the printed product to $\alpha^5 + 1$ also scores.
	$\alpha^5 + 1 = 0$ and $\alpha \neq -1$ $\Rightarrow \alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$	**A1 AG**	Uses $\alpha^5 + 1 = 0$ and states $\alpha \neq -1$ before concluding. A0 if $\alpha \neq -1$ is not stated.
		[2]	
8(d)	$\alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} = 0$	**M1**	Divides the quartic by α^2, or multiplies an expansion of the printed LHS by α^2. Condone one slip.
	$\left(\alpha + \frac{1}{\alpha}\right)^2 = \alpha^2 + 2 + \frac{1}{\alpha^2}$ $\Rightarrow \left(\alpha + \frac{1}{\alpha}\right)^2 - 2 - \left(\alpha + \frac{1}{\alpha}\right) + 1 = 0$	**M1**	Uses the expansion of $\left(\alpha + \frac{1}{\alpha}\right)^2$ to replace $\alpha^2 + \frac{1}{\alpha^2}$, or to expand the printed LHS.
	$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$	**A1 AG**	Printed result reached without error from the result of (c). Showing the printed LHS equals α^{-2} times the quartic, then citing (c), also scores.
		[3]	
8(e)	$t^2 - t - 1 = 0 \Rightarrow t = \ldots$, where $t = \alpha + \frac{1}{\alpha}$	**M1**	Solves the quadratic in $\alpha + \frac{1}{\alpha}$ by formula, completing the square or calculator.
	$t = \frac{1 \pm \sqrt{5}}{2}$	**A1**	
	$\frac{1}{\alpha} = \cos\frac{3\pi}{5} - i\sin\frac{3\pi}{5}$ $\Rightarrow 2\cos\frac{3\pi}{5} = \text{“}\frac{1\pm\sqrt{5}}{2}\text{”} \Rightarrow \cos\frac{3\pi}{5} = \ldots$	**M1**	Shows or quotes $\alpha + \frac{1}{\alpha} = 2\cos\frac{3\pi}{5}$ and equates it to their roots.

Question 8 continued

Question	Scheme	Marks	Notes
	$\cos\frac{3\pi}{5} < 0 \Rightarrow \cos\frac{3\pi}{5} = \frac{1}{4} - \frac{1}{4}\sqrt{5}$	**A1**	Positive value rejected as $\cos\frac{3\pi}{5} < 0$. In the form $a + b\sqrt{c}$, or a, b, c stated; awrt -0.309 scores A0.
		[4]	
	Total for Question 8	**12**	

Question	Scheme	Marks	Notes
9(a)	$\alpha = \frac{(6+11i)+(-8-3i)}{2}$	**M1**	Uses the midpoint of z_1 and z_2, with values substituted.
	$\alpha = \frac{-2+8i}{2} = -1+4i$	**A1 AG**	Printed result reached from the substituted midpoint without error.
		[2]	
9(b)	$z_1 - \alpha = 7+7i$ $\beta(z_1-\alpha) = \frac{(1-i)(7+7i)}{14}$	**M1**	Finds $z_1-\alpha$ and multiplies by β. Condone a slip in $z_1-\alpha$.
	$= \frac{7(1-i)(1+i)}{14} = \frac{7\times 2}{14} = 1$	**A1 AG**	Product evaluated explicitly, without error, to 1; a bare calculator result scores A0.
		[2]	
9(c)(i)	$w = e^{\frac{k\pi}{4}i}, \quad k = 0, 1, \ldots, 7$	**B1**	All eight roots, for any eight consecutive k or listed separately. $r=1$ may be omitted.
		[1]	
9(c)(ii)	$w = \beta(z-\alpha) \Rightarrow z = \alpha + \frac{w}{\beta}$	**M1**	Rearranges for z in terms of w, or otherwise rotates $z_1-\alpha$ about α through multiples of $\frac{\pi}{4}$.
	$\frac{1}{\beta} = 7(1+i)$ $z = -1+4i+7(1+i)w$	**A1**	oe
	$w = \frac{1+i}{\sqrt{2}} \Rightarrow z = -1+4i+7(1+i)\frac{1+i}{\sqrt{2}}$ $w = i \Rightarrow z = -1+4i+7(1+i)i, \ldots$	**dM1**	Substitutes at least two of the six non-real roots into their $z = \alpha + cw$, for a constant c.
	$-1+(4+7\sqrt{2})i,\ -8+11i,\ (-1-7\sqrt{2})+4i$ $-1+(4-7\sqrt{2})i,\ 6-3i,\ (-1+7\sqrt{2})+4i$	**A1**	All six, exact or awrt 3 s.f., and no others; z_1 and z_2 also listed is condoned.
		[4]	
Total for Question 9		**9**	

Question	Scheme	Marks	Notes
10(i)	$\lvert z\rvert = \sqrt{128^2 + (128\sqrt{3})^2}$ $\arg z = \pi - \arctan\sqrt{3}$	**M1**	Attempts the modulus and an argument adjusted for the second quadrant. Condone arithmetic slips.
	$-128 + 128\sqrt{3}\,i = 256e^{\frac{2\pi}{3}i}$	**A1**	
	$z_k = 256^{\frac{1}{8}} e^{\frac{1}{8}\left(\frac{2\pi}{3}+2k\pi\right)i}$ $= 2e^{\left(\frac{\pi}{12}+\frac{k\pi}{4}\right)i}$	**M1**	Takes $r^{\frac{1}{8}}$ and $\frac{\theta+2k\pi}{8}$ from their $re^{i\theta}$, for at least two values of k.
	Two of $2e^{\frac{\pi}{12}i}, 2e^{\frac{\pi}{3}i}, \ldots$	**A1**	Any two correct roots, arguments in any range.
	$2e^{\frac{\pi}{12}i}, 2e^{\frac{\pi}{3}i}, 2e^{\frac{7\pi}{12}i}, 2e^{\frac{5\pi}{6}i}$ $2e^{\frac{13\pi}{12}i}, 2e^{\frac{4\pi}{3}i}, 2e^{\frac{19\pi}{12}i}, 2e^{\frac{11\pi}{6}i}$	**A1**	All eight with $r = 2$ and arguments in $0 \le \theta < 2\pi$, and no others.
		[5]	
10(ii)			
	Shape	**B1**	Eight points equally spaced on a circle centred at O.
	Labelled point	**B1**	A to H anticlockwise, A at argument $\frac{\pi}{12}$ in the first quadrant.
		[2]	
10(iii)	$w = \frac{1}{2}\left(2e^{\frac{\pi}{3}i} + 2e^{\frac{5\pi}{6}i}\right)$	**M1**	Halves the sum of their second and fourth roots, in any form; Cartesian accepted.

Question 10 continued

Question	Scheme	Marks	Notes
	$w = e^{\frac{7\pi}{12}i}\left(e^{-\frac{\pi}{4}i} + e^{\frac{\pi}{4}i}\right)$ $\lvert w\rvert = \sqrt{2}$	**A1**	$\sqrt{2}$ or an exact equivalent.
	$\arg w = \frac{7\pi}{12}$	**A1**	Or an exact equivalent.
		[3]	
10(iv)	$w^n = (\sqrt{2})^n e^{\frac{7n\pi}{12}i}$ $\frac{7n\pi}{12} = m\pi,\ m \in \mathbb{Z}$	**M1**	Requires n times their $\arg w$ to be an integer multiple of π. Implied by $n = 12$.
	$n = 12$	**A1**	
	$a = (\sqrt{2})^{12} e^{7\pi i} = -64$	**A1**	
		[3]	
	Total for Question 10	**13**	

Question	Scheme	Marks	Notes
11(a)(i)	$\beta = -2 - 2i$	**B1**	
		[1]	
11(a)(ii)	$\alpha\beta = (-2+2i)(-2-2i) = 8$	**B1**	Or the quadratic factor $z^2 + 4z + 8$.
	"8"$\gamma = -24 \Rightarrow \gamma = \ldots$	**M1**	Sets $\alpha\beta\gamma = -24$ with their $\alpha\beta$, or divides the cubic by their quadratic factor or compares constant terms, to find γ.
	$\gamma = -3$	**A1**	
		[3]	
11(a)(iii)	$-p = -4 +$ "-3" $\Rightarrow p = \ldots$ $q = 8 +$ "-3"$(-4) \Rightarrow q = \ldots$	**M1**	Uses $\sum\alpha = -p$ or $\sum\alpha\beta = q$ with their roots, or expands their factors $(z^2 + 4z + 8)(z - \gamma)$ and compares coefficients.
	$p = 7$	**A1**	
	$q = 20$	**A1**	
		[3]	
11(b)(i)	$r = \sqrt{(-2)^2 + 2^2}$, $\theta = \pi - \arctan 1$	**M1**	Attempts $\lvert\alpha\rvert$ and an argument in the second quadrant. Not for $\pm\frac{\pi}{4}$.
	$\alpha = 2\sqrt{2}\,e^{\frac{3\pi}{4}i}$	**A1**	$r = 2\sqrt{2}$, $\sqrt{8}$ or awrt 2.83 and $\theta = \frac{3\pi}{4}$ or awrt 2.36, in the form $re^{i\theta}$; not for r and θ stated alone.
		[2]	
11(b)(ii)	$(-2+2i)^n = \left(2\sqrt{2}\,e^{\frac{3\pi}{4}i}\right)^n$ $= (2\sqrt{2})^n e^{\frac{3n\pi}{4}i}$	**M1**	Applies $(re^{i\theta})^n = r^n e^{in\theta}$, or de Moivre, to their α in modulus-argument form from (b)(i).
	$= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + i\sin\frac{3n\pi}{4}\right)$	**A1 AG**	Correct α^n from $2\sqrt{2}\,e^{\frac{3\pi}{4}i}$ leading without error to the printed result; combined steps accepted.
		[2]	
11(b)(iii)	$\beta^n = \left(2\sqrt{2}\,e^{-\frac{3\pi}{4}i}\right)^n$ $= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} - i\sin\frac{3n\pi}{4}\right)$	**M1**	Raises β in exponential or modulus-argument form, argument $-\frac{3\pi}{4}$ oe, to the power n, or takes β^n as the conjugate of α^n.
	$\alpha^n + \beta^n = 2(2\sqrt{2})^n \cos\frac{3n\pi}{4}$	**A1**	

Question 11 continued

Question	Scheme	Marks	Notes
	$\gamma^n = (-3)^n$ $\alpha^n + \beta^n + \gamma^n$ $= 2(2\sqrt{2})^n \cos\frac{3n\pi}{4} + (-3)^n$	**A1 AG**	Adds $(-3)^n$ from $\gamma = -3$ to a correctly obtained $\alpha^n + \beta^n$, reaching the printed result without error.
		[3]	
	Total for Question 11	**14**	

Question	Scheme	Marks	Notes
12(a)	$-729 = 3^6 e^{(\pi+2k\pi)i}$ $\Rightarrow z = 3e^{\left(\frac{\pi}{6}+\frac{k\pi}{3}\right)i}$	**B1**	Modulus 3 and arguments $\frac{\pi}{6}+\frac{k\pi}{3}$, soi by at least two correct roots.
	$A: 3e^{\frac{\pi}{6}i}$, $B: 3e^{\frac{\pi}{2}i}$, $C: 3e^{\frac{5\pi}{6}i}$ $D: 3e^{\frac{7\pi}{6}i}$, $E: 3e^{\frac{3\pi}{2}i}$, $F: 3e^{\frac{11\pi}{6}i}$	**B1**	All six in exponential form; arguments may differ by multiples of 2π, such as $-\frac{\pi}{6}$ for F.
		[2]	
12(b)	$z_G = \frac{z_A+z_B}{3}$, $w = \frac{z_G}{z_A} = \frac{z_A+z_B}{3z_A}$	**M1**	Forms a centroid as the mean of its three vertices, with O as 0, and divides it by the corresponding root.
	$z_B = z_A e^{\frac{\pi}{3}i} \Rightarrow w = \frac{1+e^{\frac{\pi}{3}i}}{3}$	**A1**	Any correct expression for w, oe such as $\frac{1}{2}+\frac{\sqrt{3}}{6}i$.
	$1+e^{\frac{\pi}{3}i} = \frac{3}{2}+\frac{\sqrt{3}}{2}i$ $\Rightarrow r = \ldots\ [= \sqrt{3}]$, $\theta = \ldots\ [= \frac{\pi}{6}]$	**M1**	Finds the modulus and argument of their w or its numerator, by any method such as Cartesian form or factorising out $e^{\frac{\pi}{6}i}$. Condone slips.
	$w = \frac{1}{\sqrt{3}}e^{\frac{\pi}{6}i}$	**A1**	Exponential form required; oe such as $\frac{\sqrt{3}}{3}e^{\frac{\pi}{6}i}$, or awrt 0.577 and 0.524.
		[4]	
12(c)	$p = \text{“}w\text{”}^6 \times (-729)$ or $p = z_G^6 = \left(\sqrt{3}\,e^{\frac{\pi}{3}i}\right)^6$	**M1**	Multiplies the sixth power of their w by -729, or finds the sixth power of one of their points G to L.
	$p = 27$	**A1**	
		[2]	
	Total for Question 12	**8**	

Question	Scheme	Marks	Notes
13(i)	$w = e^{\frac{(2k+1)\pi}{11}i}$ $k = 0, 1, \ldots, 10$	**B1 B1**	**B1** Any correct general form, such as modulus-argument; k need not be restricted. **B1** Exactly eleven distinct roots, by a range of k such as -5 to 5, or listed.
		[2]	
13(ii)	$\left(\frac{z+1}{z-1}\right)^{11} = -1$ $\Rightarrow \frac{z+1}{z-1} = w = e^{i\theta}$, $\theta = \frac{(2k+1)\pi}{11}$	**M1**	Divides by $(z-1)^{11}$ and sets $\frac{z+1}{z-1}$ equal to their roots from (i). Condone $z \neq 1$ not stated.
	$w(z-1) = z+1 \Rightarrow z = \frac{w+1}{w-1}$	**M1**	Rearranges to make z the subject. Condone sign slips.
	$z = \frac{e^{i\theta/2}+e^{-i\theta/2}}{e^{i\theta/2}-e^{-i\theta/2}}$ $= \frac{2\cos(\theta/2)}{2i\sin(\theta/2)}$	**dM1**	Multiplies by $e^{-i\theta/2}$, or uses half-angle identities on $1 \pm \cos\theta$ and $\sin\theta$, reaching a ratio of trigonometric functions of $\frac{\theta}{2}$.
	$z = -i\cot\left(\frac{(2k+1)\pi}{22}\right)$	**A1**	cso, with $\frac{1}{i} = -i$; any equivalent set of eleven values.
	z real $\Rightarrow w$ real $\Rightarrow w = -1 \Rightarrow z = 0$ One real root ($m = 11$), so ten non-real	**E1**	Identifies $z = 0$ as the only real root, by w real or by $\cot\left(\frac{m\pi}{22}\right) \neq 0$ for $m \neq 11$, and concludes ten non-real roots.
	$-i\cot\left(\pi - \frac{r\pi}{22}\right) = i\cot\left(\frac{r\pi}{22}\right)$ $= -i\cot\left(-\frac{r\pi}{22}\right)$ $m = \pm1, \pm3, \pm5, \pm7, \pm9$	**A1 AG**	Maps $m = 13, \ldots, 21$ to $-9, \ldots, -1$ by $\cot(\pi - x) = -\cot x$ with cot odd, or by period π, and states the printed set.
		[6]	
13(iii)	$(-i \times i)\cot^2\left(\frac{m\pi}{22}\right) = -i^2\cot^2\left(\frac{m\pi}{22}\right)$ $= \cot^2\left(\frac{m\pi}{22}\right)$	**B1**	Needs $i^2 = -1$ used explicitly.
		[1]	
13(iv)	$\cot^2\left(\frac{\pi}{22}\right)\cot^2\left(\frac{3\pi}{22}\right)\cdots\cot^2\left(\frac{9\pi}{22}\right) = 11$	**M1**	Pairs the roots for m and $-m$ and uses (iii) to equate the product of the five $\cot^2$ values to 11.
	$\frac{1}{11}$	**A1**	$\frac{1}{11}$ or awrt 0.0909.
		[2]	
	Total for Question 13	**11**	